THIS STAMP, ON THIS PROJECT
APPLIES ONLY TO SECTIONS AUTHORED BY:

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SECTION 26 00 01

GENERAL ELECTRICAL PROVISIONS

PART 1 - GENERAL

Products under this contract must meet minimum specifications requirements in detail without exception unless specifically noted and approved as provided in these Specifications. Equipment submitted for review must clearly state on cover sheet any differences from specified product. Equipment substitution or submittal review does not relieve Contractor from meeting all requirements of specified item.

1.01 DEFINITIONS

A. Definitions herein are intended as advisory and shall not limit requirements within the Contract Documents. Where a conflict of definitions exists, the more stringent standard shall be used. Where a term is defined on a Drawing the Drawing definition shall be used for that drawing. Not all definitions are included. Trade standard terms are not defined.

1.02 CONTRACT DOCUMENTS

A. The Contract Documents are inclusive. All requirements of all Contract Documents shall be binding as if repeated herein and within this Division as required by any other Division or Contract Document.

B. This Division does not express or imply separation of the Contract Documents and shall not be considered as separation of the Work.

C. See Advertisement For Bids, Instructions to Bidders, Supplemental Instructions to Bidders, General Conditions, Supplemental General Conditions, Drawings and Specifications, and modifications incorporated in the documents before execution of the Agreement.

D. Conflicts: If any conflicts exist the more stringent is required.

1.03 SCOPE OF WORK

A. General: Provide complete and functional electrical systems as specified, as shown on Drawings, as required, and as intended. Work generally includes, inspections, electrical distribution, lighting, devices, wiring systems, control cabling, raceways, and control systems.

B. Omissions: Contractor shall be responsible for additional labor, or additional material necessary for the proper execution of the Work. Omissions of expressed reference to any item shall not relieve the responsibly to conform to the Contract Documents

C. Scope of Electrical Work

1. Provide per current California Electrical Code (CEC).

2. Materials and workmanship shall be furnished for complete, tested, and operating electrical systems as shown on the drawings and specified herein.

3. Electrical work is to include the electrical service. Complete to the point of connection with the serving utility. Any changes of or work required by the serving utility, are part of this work and shall be fully included in the bid price.

4. Work is also to include main distribution panel, feeder system and branch circuit panels. Complete branch circuit wiring. Light fixtures, wall switches, receptacles and similar items, exterior lighting and wiring, and wiring and connection to all mechanical equipment, all fire alarm, phone system, unit phone wires, and kitchen equipment as required.

5. Provide Power Connections to Equipment provided by others, specified not shown, and/or part of a complete and operable system. This equipment may, or may not be shown, and generally includes, and is not limited to:

    a. Fire Alarm Control Panel (FACP), Fire Alarm Equipment Power Supplies, etc.

    b. Lighting Control Panels, Lighting Controller Equipment, Lighting Controls.

    c. Building Management Control System, Direct Digital Controls, HVAC Controls, and the like.

    d. Automatic Door Operators, ADA Doors, Sliding Automatic Doors, and the like.

    e. Unitary, generally self-contained access controls.

    f. Doorbells, door cameras, door locks, door strobes/flashers, etc.

    g. Sump Pumps, freeze protection, and similar site equipment (including equipment located within the building footprint).

6. Control Connections to unitary Equipment provided by others, specified, not shown, and/or part of a complete and operable system. This equipment may, or may not be shown, and generally includes, and is not limited to, control wiring for:

    a. Automatic Door Operators, ADA Doors, Sliding Automatic Doors, and the like.

    b. Unitary, generally self-contained access controls

c. Doorbells, door cameras, door locks, door strobes/flashers, etc.

d. Sump Pump Controls, alarms, and the like.

e. Water Level/Monitoring alarms, controls, etc.

7. The lists above are intended to indicate equipment that is generally shown/specified elsewhere. Where specified, and/or shown otherwise, and/or specified by electrical follow additional power, and control requirements as indicated.

8. Provide Equipment per Manufacture's Operation, Installation, and Programming requirements. Follow manufacture's Guidelines, Installation manuals. Provide additional equipment and as required for a complete, and satisfactory Installation.

1.04 CONFORMANCE WITH REQUIREMENTS

A. General: All Work shall conform to the reasonable requirements of the project within the scope of the project and authorizations. All work shall conform to the methods and requirements of Code at the location of the Work.

1. Access and inspection: All portions of the Work shall be accessible to inspections and review at all reasonable times during construction. Contractor is responsible for providing access for review and inspection of the Work. Contractor shall secure written inspection reports prior to concealing Work. Contractor is responsible for damages to properly review the Work due to lack of at least 7 Days advance written notification to the Architect, and Engineer that Work is ready for inspection.

2. Plenums: Provide raceway where conductors, and/or cables are routed through plenums. If specifically noted, or specified raceway is not required, plenum routes may utilize conductors/cables rated for use in plenums, and shall be marked as such per NEC. Supporting devices utilized shall also be rated for use in plenums.

B. Accounting: Provide general accounting information as to labor and equipment costs to assist in determination of modifications to the Contract. Provide accounting breakdown when required for securing Owner financing, or for analysis of equipment costs or equipment payback periods, as well as information for Owner incentives.

1.05 COORDINATION OF TRADES

A. Check all other trade drawings to avert potential installation conflicts. Should major changes from the Drawings be required to resolve potential conflicts, notify the Architect and secure written approval and agreement on necessary adjustments prior to start of installation.

B. Check all equipment locations and connections on the site for coordination with other Divisions equipment and connections and structure and the like.

C. Contractor is responsible for scheduling trades to properly execute all the Work as intended.

1.06 STANDARD OF CARE AND QUALIFICATIONS

A. General: Contractor shall be experienced and knowledgeable to Provide Work. Owner is not responsible for improper operation, incompliance, or installation due to Contractor's lack of knowledge or experience. Upon request, and where requested herein the Contractor shall supply qualifications and experience. Drawings are presented with industry terms, statements, and trade practices and it is the responsibility of the Contractor to be familiar. Provide written notification prior to Bid to the Architect if any representation is not understood, or outside standard practice.

B. Like Materials and Quality Control: All systems provided shall be new and of like materials provided through manufacturer authorized distributors. Provide equipment of same system and type by same manufacturer. Items of the same by different manufacturers will be rejected. Equipment shall conform to all applicable Code and applicable listing criteria as of the date of the Contract Documents. Equipment determined to be manufactured under any other listing or Code prior to the date of the Contract is not acceptable, even if the equipment is new or has not been used. All equipment provided to project shall be listed by an approved listing organization.

1.07 EXAMINATION OF SITE

A. Examine Site of Work prior to making Bid. Ascertain all related physical conditions.

B. Verify at the Site of Work prior to Bid scale dimensions shown due to exact locations, distances, and levels will be governed by actual field conditions.

C. Owner will not be responsible for any loss or costs that may be incurred due to a Bidder's failure to fully inform themselves prior to Bid in regard to conditions pertaining to the Work and nature of the Work.

1.08 SPARE RACEWAY

A. In addition to spare raceways required herein (see Panelboards), provide spare raceways emanating from equipment pads where pads prevent access to equipment mounted on pads. Examples include transformers, MCC's, switchboards, generators, and the like. Route spares to landscaped areas at exterior locations or to equipment room interior accessible walls at interior locations.

B. Provide not less than (2) 4” and (4) 2” spare raceway. Coordinate termination and route locations/direction with Owner. See Raceway Specifications. Additional Spare Raceway may be shown on Drawings, or required elsewhere herein.

1.09 MINOR DEVIATIONS

A. Make minor changes in equipment locations and equipment connections as directed or required without extra cost.

1.10 RECORD DRAWINGS

A. Maintain a marked set of prints at job site at all times. Show all changes from the original drawing set whether visible or concealed. Include all addendums, field orders, change orders, clarifications, request for information drawn responses, and deviations. Dimension accurately from building lines, floor, or curb elevations. Show exact location, elevation, and size of conduit/raceway, access panels and doors, equipment, and all other information pertinent to the Work.

B. At project completion, submit marked set to Architect for review.

1.11 TRAINING

A. Provide training of Owner’s selected staff for all electrical systems specified herein.

B. Training of systems shall be digitally recorded, and two copies shall be distributed to Owner.

C. Notify and Coordinate with Owner for training and attendance not later than 15 Days prior to training.

D. Provide 1 hour of general system training in addition to training indicated below.

1. VFD Updated Controls- 6 hour

E. Training shall be conducted by qualified individuals familiar with the Work, and with the equipment.

F. Instructor shall be familiar with programming and operation of equipment and shall provide instruction to do such.

G. Provide contact information to Owner for an additional 1 hours support for all electrical systems.

H. Training shall not occur prior to systems being fully inspected, operational, and complete.

I. Utilize necessary training materials, conduct training at project location including walk-through of equipment on-site.

J. Provide Owner with all required Operation, Maintenance, and Programming manuals provided by equipment manufacturer.

K. Owner shall determine attendee's at training, not the contractor. Contractor shall re-train if attendee's were not selected by the Owner.

1.12 WARRANTY

A. Warrant Work, materials, and equipment for not less than one year.

B. Provide additional warranty as required herein.

1.13 PRODUCTS

THIS PART NOT USED

1.14 EXECUTION

THIS PART NOT USED

END OF SECTION

SECTION 26 00 20

ELECTRICAL DEMOLITION

PART 1 - GENERAL

1.01 INTENT

A. The electrical system shall be adjusted to accommodate installation of the new work and is required to meet Code. Necessary adjustments are shown on Drawings and indicated herein.

1.02 EXISTING CONDITIONS

A. The locations of existing utilities and equipment are shown in an approximate way only and have not been independently verified by the Owner or Owner's Representative. Contractor shall determine the exact location of every existing utility before commencing work, and agrees to be fully responsible for any and all damages which might be occasioned by the Contractor's failure to exactly locate and preserve utilities and equipment. Replace damaged items with new material to match existing. Promptly notify Owner if utilities are found which are not shown.

B. Existing equipment to be removed may contain PCB's, Mercury, and other hazardous materials.

1.03 PRODUCTS

THIS PART NOT USED

PART 2 - EXECUTIONS

2.01 DEMOLITION

A. Only partial existing electrical shown. Locations of items shown on the Drawings as existing are partially based on field inspection of unconcealed equipment and record drawings that may contain errors. The contractor shall verify the accuracy of the information shown prior to bidding and provide such labor and material as is necessary to accomplish the intent of the Contract Documents.

B. Remove existing fixtures, and associated electrical equipment from areas scheduled for remodeling, relocation, or demolition unless indicated as retained, or noted on the Drawings.

C. Maintain continuity of existing systems that remain. Remove or relocate electrical boxes, conduit, wiring, equipment, fixtures, etc. as may be encountered in removed or remodeled areas in the existing affected by this work. Wiring which serves usable existing equipment shall be removed and restored clear of the construction or demolition. If existing junction boxes will be made inaccessible, or if abandoned

outlets serve as feed through boxes for other existing electrical equipment which is being retained, new conduit and wire shall be provided to bypass the abandoned equipment. If existing conduits pass through areas being removed or remodeled, new conduit and wire shall be provided to reroute clear of the construction or demolition and maintain service to the existing load.

D. Remove abandoned wiring, and leave site clean.

E. Do not disconnect service without approval from the Owner and Utility. Coordinate demolition work with the utility.

F. Verify with the General Contractor a location for storage of materials, supplies, tools, rubbish, etc. prior to start of Work.

G. Include disposal costs in bid unless item is specifically noted as salvage to Owner, or retained. Include removal, retainment, and salvage to Owner costs in bid; including relocating equipment to another location on the Owner's site. Owner has first right of refusal for any other equipment scheduled for demolition not already noted as salvaged to Owner. Owner may elect to salvage any equipment scheduled for disposal.

H. Protect equipment salvaged to Owner. For equipment with disposal/recycle/salvage value, such proceeds shall be passed to Owner through inclusion in total bid.

I. Equipment being retained shall be protected from damage, stored, labeled and reused. Contractor is responsible for damage to retained equipment. Contractor is responsible for labeling retained equipment with non-damaging labels. Should Contractor not label retained equipment, labels become separated, and/or labeling is inaccurate, Contractor shall be responsible for costs to determine equipment type, labeling, as part of a survey to re-install retained equipment in a specific condition, location, with specific required function. Equipment varies and labeling/function type of retained equipment is essential to protect success, schedule, and budget. Contractor is responsible for schedule delays, and equipment costs, as a result of equipment damage, replacement equipment procurement, (due to Contractor's damage or not labeling properly), and/or labeling errors, omission, adherent.

END OF SECTION

SECTION 26 00 26

SUBMITTALS AND SHOP DRAWINGS

PART 1 - GENERAL

1.01 REQUIREMENTS

A. Refer to Division 1

B. Organization

1. Provide 3-ring type hard cover notebook with 3-hole punch product data sheets.

2. Order submittals in logical form with tab dividers indicating specification section, and specification title

3. Equipment shown on schedules shall be in logical order as the equipment appears on the schedule (i.e. light fixture type A precedes light fixture type Z), and be submitted with IDs matching schedule(s).

4. Submit 5 copies for review. Not all copies will be returned to Contractor.

5. Clearly readable electronic submittals are permitted in – Lieu of initial hard copies provided they are printed by the Contractor with the O&M Manuals for Owner's hard-copy.

    a. Contractor is responsible to verify receipt of electronic submittals by Engineer.

    b. Electronic Submittals shall include the project title in the subject line with unique submittal number, and description of submittal.

    c. Submit to Email address: submittals@fluentengineering.com

        1) Fluent Engineering has no limit on E-mail sizes; however, from time to time our email service provider may restrict incoming e-mails beyond our control.

        2) Contractor is responsible for any damage as a result of viruses or other malicious software or links to such contained with submittal

        3) Fluent Engineering will accept only PDF electronic files. ZIP files, links, images files, images within an email message, and Dropbox type services, etc. will not be opened/accepted.

4) Open file sharing programs such as Dropbox are not permitted and are considered a compromise of project security / non-disclosure agreements.

5) Coordinate with Engineer if PDF E-mail is too large to be sent.

6) Engineer may elect on a case by case basis to utilize the/a Contractor's online submittal system with Owner's approval.

7) Otherwise, use hardcopy format.

C. Allow no less than 20 Days for review by Engineer.

D. Contractor is responsible to submit and verify receipt of comments for all submittals.

E. Resubmittals shall contain all items included in pervious submittals with changes clearly identified with a cover letter listing the changed items, or if permitted by Division 1 only resubmit items that have changed with changes clearly identified. Only revised items will be reviewed.

F. No item requiring review shall be delivered to the site or otherwise provided to the Project until submittals have been reviewed by the Engineer.

1.02 DEFINITIONS

A. Manufacturing Data: Information regarding the product(s) and equipment issued by the manufacturer as described below.

B. Manufacturer's Label: Manufacturer's label shall include a typewritten list of manufacturer's name, sizes and model or catalog numbers.

C. Manufacturer's Catalog Data: Manufacturer's catalog data shall include standard catalog information (Cut Sheets) marked to indicate specific equipment and options for complete and functional system. All components of the system shall be included. Include listing information. Include installation instructions.

D. Manufacturer's Technical and Engineering Data: Manufacturer's technical and engineering data shall include materials, dimensions, details, installation instructions, weights, capacities, illustrations, wiring diagrams, control diagrams, control schematics, piping diagrams, connection diagrams, performance data, trip curves, listings, mix design, test results, and any other information required for a complete evaluation of the equipment specified, and to verify compliance with the Contract Documents. All available details shall be included with any modifications to the equipment indicated. All manufacturers and associated model numbers used for complete system shall be indicated.

E. Shop Drawings: Shop drawings are Construction drawings of items manufactured specifically for this project. Shop drawings shall include dimensions, construction details, weights, and additional information to identify the physical features of the system or piece of equipment. Drawings shall be adequately sized and scaled for a complete review.

F. Samples: Samples include actual example of the equipment to be installed. Include actual color, finish, and functioning replica of equipment to be installed. Samples will be returned to the Contractor when submitted with pre-paid postage.

G. Certifications and Qualifications: Submit list of past projects with same systems. Submit information listing references, copies of certificates issued by manufacturer, school, and standards organizations. Submit information mandated in specific specification section.

1.03 SUBMITTALS REQUIRED

A. Product Evaluation Data. 5 copies of product literature. The submittal schedule for product evaluation data is as indicated below. Each item requiring a submittal is given the following code:

L	Manufacturer's Label
C	Manufacturer's catalog data (Cuts)
E	Manufacturer's technical and engineering data
S	Shop drawings
SA	Samples
CR	Certifications
Q	Qualifications

1.04 SUBMITTAL SCHEDULE

Division 26 - Electrical

Section 26 05 19- BUILDING WIRE AND CABLES ……...…….C
Section 26 05 26- GROUNDING ……..…….....C
Section 26 05 29- SUPPORTING DEVICES …….…….L
Section 26 05 33- RACEWAYS & FITTINGS ……..….....L
Section 26 05 33.16- OUTLET, JUNCTION, AND PULL BOXES ……..……..C
Section 26 05 53- ELECTRICAL IDENTIFICATION …….L,SA
Section 26 05 60- OVERCURRENT PROTECTIVE DEVICES …….……C,E,S
Section 26 05 83- WIRE CONNECTIONS ……...……..L
Section 26 27 26- WIRING DEVICES ……....….C,E,S
Section 26 29 12- DISCONNECTS & MANUAL MOTOR STARTER ……...……..C,S
Section 27 05 01- CONTROL CABLING ……...…….C, S

PART 2 - PRODUCTS

THIS PART NOT USED

PART 3 - EXECUTION

THIS PART NOT USED

END OF SECTION

SECTION 26 05 19

BUILDING WIRE & CABLES

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Wires and Cables

1.02 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

1.03 DELIVERY, STORAGE AND HANDLING

A. Deliver new wire and new cable to site in new packaging with standard cable coils/reels. Packaging shall clearly show length, wire size, wire/cable type, and manufacturer.

B. Protect products from weather, moisture, and damage.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Building Wiring & Insulation: Copper, 98 percent conductivity, stranded. Solid may be used at contractor's option for wire smaller than #8 AWG. 600 Volt insulation, Type THHN for dry interior and damp interior locations. Type THW, THWN or XHHW for wet locations, and exterior locations.

B. Conductor cable with conductors smaller than #12 AWG for branch circuits not permitted.

C. Exterior cables exposed to sunlight shall be listed "sunlight resistant."

D. Control panel wiring no smaller than #14 AWG stranded switchboard Type MTW unless otherwise specified on the Drawings or required by system manufacturer.

E. Motor control wires shall be no smaller than #14 AWG.

F. Wire for other areas as shown on the Drawings.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Parallel feeders shall have identical conductor length.

B. Use UL listed pulling lubricant for greater than equivalent #4 AWG wire diameter.

C. Use UL listed pulling lubricant for pulls greater than 75 feet.

D. Remove moisture from raceway prior to wire pull.

E. Provide copper grounding conductors. Provide a ground wire through conduits. Utilize the ground wire as the equipment grounding conductor no smaller than #12 AWG otherwise sized as shown and per NEC.

F. Do not splice feeders, or services. Splices only permitted in accessible junction or outlet boxes where circuit routes deviate. Do not splice or tap branch circuits terminating in a single outlet.

G. Color code conductors per NEC to designate neutral, phase, and ground as follows:

CONDUCTOR	120/208
Phase A	Black, or per (E) facility standard
Phase B	Red, or per (E) facility standard
Phase C	Blue, or per (E) facility standard
Neutral	White
Ground	Green
Travelers	Pink, or per (E) facility standard

CONDUCTOR	480/277
Phase A	Brown, or per (E) facility standard
Phase B	Purple, or per (E) facility standard
Phase C	Yellow, or per (E) facility standard
Neutral	White
Ground	Green
Travelers	Pink, or per (E) facility standard

H. Wires shall be factory color coded. Coloring shall be integral to the insulation. Plastic tape permitted on #6 AWG and larger where insulation coloring is not available or practical. Apply tape in spiral half-lap over exposed portions of cable at all locations that cable is accessible.

I. Conductors shall be identified with circuit number where conductors are accessible such as at terminals, outlets, switches, circuit breakers, motor control centers, etc. Identify the ends of a given conductor circuit the same.

J. Do not install wires of different voltage systems in same raceway, box, or other enclosure. Control voltage is permitted in same enclosure only where specific equipment is listed for multiple voltage use, and a listed voltage barrier is provided.

K. Radius of cable bends shall not be less than 10 times the outer diameter of the cable.

L. Do not install cable within conduit per NEC.

M. Follow standards of practice for storage, handling, and termination of aluminum conductors. Provide anti-oxidation gel, and remove any oxidation by approved means when terminating conductors.

N. Route parallel or perpendicular to building lines.

END OF SECTION

SECTION 26 05 26

GROUNDING

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Electrical systems grounding.

B. Signal systems grounding.

1.02 APPLICABLE STANDARDS

A. Underwriters Laboratories (UL)

1. UL 467 Standard for Grounding and Bonding Equipment

B. Institute of Electrical and Electronic Engineers (IEEE)

1. IEEE 81 Guide for Measuring Earth Resistively, Ground Impedance, and Earth Surface Potentials of a Ground System Part 1: Normal Measurements

2. IEEE 142 Recommended Practice for Grounding of Industrial and Commercial Power Systems

1.03 APPLICABLE REGULATIONS

A. National Fire Protection Association (NFPA)

1. NFPA 70 National Electrical Code (NEC)

B. NEC references below are based on the 2023 edition. Contractor shall meet current NEC requirements.

1.04 TESTS

A. Measure ground grid resistance with earth test megger and provide additional listed and approved earth grounding devices and conductors as required until resistance is 25 ohms or less. Inform engineer in writing, if resistance is greater than 5 ohms.

PART 2 - PRODUCTS

2.01 GROUNDING ELECTRODES

A. Encased Electrode: NEC 250.52(A). One, no smaller than #4 bare solid copper conductor. Install in concrete foundation or footing near contact with earth. Connect to steel reinforcing bars, where available, not less than two times.

B. Ground Rods: 3/4" diameter, 8' long, copper, with approved clamp near surface. Listed as ground rod for direct contact with earth.

C. Grounding Electrode: NEC 250.52(A).

2.02 GROUNDING CONDUCTORS, AND JUMPERS

A. Size: Per NEC 250.

B. Material: Copper.

C. Protection: Conductors not in raceway or concealed shall be insulated. Provide raceway where shown or required for physical protection.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Regardless of TESTS results above, Provide not less than the following:

1. Underground Metal Cold Water Pipe electrode; If underground Metal Cold Water Pipe is not available Provide Ground Ring.

2. Encased Electrode.

3. Two (2) Ground Rods or Ground Rods as shown.

4. Connection to building steel when present.

B. Grounding Electrodes: Bond all electrodes together. Do not provide other type of electrode than shown without written approval. Provide additional quantity of electrodes as required by TESTS above.

C. Provide access to all grounding electrode conductor connections.

D. Electrically bypass water meters when required by utility with use of full-size bonding jumper rated for location installed and with pipe clamps routed around and clear of meter.

3.02 POWER AND SIGNAL SYSTEM GROUNDING

A. All equipment grounding conductors shall be routed through same equipment conductor raceway from beginning to end (distribution source to load).

B. Every metallic raceway (including MC Cable, and :Liquidtight Flexible Metal Conduit) circuit/feeder is intended to have dual ground paths from the raceway/sheath itself, and the separate equipment grounding conductor within the raceway/cable.

C. Metallic raceways are not approved as equipment grounds.

D. Circuit Grounding: Install grounding bushings, studs, and jumpers at distribution centers, pullboxes, motor control centers, panelboards, and junction boxes.

E. Ground Connections: Clean surfaces thoroughly before applying ground lugs or clamps. If surface is coated, the coating must be removed down to the conductive material. After the coating has been removed, apply a listed and approved noncorrosive compound to cleaned surface and connections. Where galvanizing is removed from metal, it shall be re-applied or painted.

F. Bonding Jumpers: Provide with green insulation and size not smaller than per NEC and larger where shown. Connection to neutral only at service neutral bar. Bonding jumpers shall be contiguous without break, joint, or splice.

G. Service Panel:

1. Connect the various feeder green grounding conductors to the ground bus in the enclosure with suitable pressure connectors.

2. Connect the grounding electrode conductor to the ground bus.

3. Connect the neutral to the ground bus as the main bonding jumper.

4. Connect metallic conduits, which terminate without mechanical conductive connection to the enclosure, by grounding bushings and ground wire to the ground bus.

H. Feeders and Branch Circuits: Install green grounding conductors with feeders and branch circuits. Additional locations and systems as shown.

I. Raceway Systems:

1. Ground all metallic enclosed raceway systems.

2. All enclosed raceway connecting to equipment shall contain a grounding conductor.

3. Conduit systems shall contain a grounding conductor.

4. Bond grounding conductor at beginning and end of raceway provided for mechanical protection containing only a grounding conductor.

J. Boxes, Cabinets, Enclosures, and Panelboards:

1. Bond the grounding wires to each pullbox, junction box, outlet box, cabinets, and other enclosures through which the ground conductors pass.

2. Provide lugs in each box and enclosure for ground wire termination.

3. Provide ground bars in panelboards, bolted to the housing, with sufficient lugs for terminating the ground wires.

K. Receptacles - Refer to Section 26 27 26.

L. Ground lighting fixtures to the equipment grounding conductor of the wiring system.

M. Fixed electrical equipment shall have a ground lug installed for termination of the equipment ground conductor.

N. Motors: Install a separate insulated equipment grounding conductor from the equipment ground connection in the motor controller through the raceway and flexible conduit to the ground terminal on the motor housing. Ground motor controller through feeder raceway. No reductions.

O. Variable Frequency (Speed) Drives (VFD/VSD): Additional grounding requirements as shown on Drawings, and with not less than 3 concentric Ground conductor cable between VFD and motor.

P. Control and Signaling Equipment: Ground metallic enclosures and raceways, terminate shields and drain wires to building ground system. Provide additional grounding as required by equipment manufacturer.

3.03 TESTING

A. Test per IEEE 81.

B. Grounding Electrode Conductor:

1. Measure resistance between switchboard ground bus and each grounding electrode, using a Megger and a single length of additional wire.

2. Measure resistance between both ends of the additional wire used.

3. Grounding Electrode Conductor resistance is the difference between 1 and 2.

4. Correct any inadequate connections as indicated.

END OF SECTION

SECTION 26 05 29

SUPPORTING DEVICES

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Raceway Supports.

B. Cable supports.

C. Provide all hardware and materials to support, as required, a complete and congruent raceway system.

1.02 APPLICABLE STANDARDS

A. National Fire Protection Association (NFPA)

1. NFPA 70 National Electrical Code

B. Underwriters Laboratories (UL)

1. UL 2239 Hardware for the Support of Conduit, Tubing, and Cable

C. National Electrical Contractors Association (NECA)

1. ECA 101 Standard for Installing Steel Conduit

PART 2 - PRODUCTS

2.01 RACEWAY SUPPORTS

A. Single Runs: Steel rod hangers, galvanized single hole conduit straps, or ring bolt type hangers with spring clips. Adhesives, tape, staples, zip/wire ties, or "J-nails" not acceptable.

B. Multiple Runs: Rack with 25 percent spare capacity. Maximum width per manufacturer's recommendations.

C. Vertical Runs: U-channel support with conduit fittings.

D. All hardware such as inserts, straps, bolts, nuts, screws and washers shall be galvanized or plated steel.

E. PVC coated galvanized steel in exterior and wet locations.

F. Channel manufacturers: Kindorf, Unistrut, or approved.

2.02 CABLE SUPPORTS

A. Approved plastic coated wire-ties.

B. Approved PVC coated hangers.

C. Building studs as permitted by Code and specifications.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Supporting devices shall be listed for the location installed. Supports shall be of like material of raceway and be rated for location installed.

B. Layout to maintain headroom, neat mechanical appearance, and to support equipment loads required.

C. Exact location and spacing between supports per manufacturer's recommendations and NEC requirements.

D. Provide adequate spacing to prevent moisture build-up. All runs of conduit shall be arranged so as to be devoid of traps wherever possible.

E. Cable “Sag” greater than 3-Inches from valley to peak of run, not acceptable.

END OF SECTION

SECTION 26 05 33

RACEWAY & FITTINGS

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Conduit, Fittings, and Tubing.

B. Flexible Conduit.

1.02 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code--Chapter 3.
2. Underwriters Laboratories (UL) 6,
3. UL797
4. UL1990

PART 2 - PRODUCTS

2.01 MATERIALS AND COMPONENTS

A. General: No smaller than ¾-inch unless otherwise shown or indicated herein.

B. Conduit and Tubing: Electrical metallic tubing, galvanized rigid steel threaded conduit, Schedule 40 PVC.

C. Flexible Conduit: Flexible plastic jacketed type with liquidtight connectors and steel wrap armor (liquidtight flexible metallic conduit).

D. Fittings:

1. General: Listed and approved for purpose. Water, gas, concrete tight where required.
2. Electrical Metallic Tubing (EMT): Connectors to be steel. All connectors shall have factory insulated throats. Couplers and connectors shall be compression, setscrew type.
3. Galvanized Rigid Steel Conduit (GRC): Threaded. Do not use pressure type. Provide factory insulated throats on bushings.

4. PV Coated Galvanized Rigid Steel Conduit (PGRC): Same as GRC, with PVC coating meeting UL White Book Guidance, Steel Tube Institute.

5. Liquidtight Flexible Metallic Conduit: Continuous copper ground in core; approved watertight.

E. Expansion Joints: Offset or sliding type with bending straps and clamps. Listed for purpose.

F. Entrance Seal: Shall be modular, mechanical type, consisting of inter-locking synthetic rubber links shaped to continuously fill the annular space between the pipe and the wall opening. The elastomeric element shall be sized and selected per manufacturer's recommendations. Garlock Link-Seal or approved.

G. Underground Marking Tape: 6-inches wide, yellow, low-density polyethylene 4 mil thickness. Imprinted: "CAUTION: STOP DIGGING - BURIED ELECTRIC LINE BELOW" and current date. Tape for telephone line similar, except green.

H. Conduit seal: Shall be modular, mechanical type. Listed for use., concentric cable/conductor seals on inside of raceway. Waterproof, Roxtec or approved system

2.02 TYPE

A. Utilize PGRC in concrete with concrete-tight connectors.

B. Utilize GRC for exterior with watertight connectors.

C. Utilize electrical metallic tubing concealed in finished interior spaces.

D. Utilize electrical metallic tubing exposed in unfinished spaces, where not subject to physical damage.

E. For underground conduit, utilize Schedule 40 PVC or PGRC. Provide PGRC elbows and PGRC risers through penetrations where PVC is used.

F. Utilize surface metal raceways for exposed runs in finished areas. Paint to match wall finish. Use only where shown on Drawings or where approved.

G. Connections to motors, vibrating equipment, and movable equipment shall be with flexible metallic conduit or liquidtight flexible metallic conduit. Use liquidtight type in damp locations. No smaller than 1/2-inch for motor connections. Use 3/8-inch only for light fixture wiring where provided by light fixture manufacturer. Provide sufficient length of flexible conduit to stop vibration into connecting support. Sizes not noted on the Drawings shall be as required by the NEC and no smaller than upstream connection conduit size.

H. Flexible metallic conduit not to exceed 6-feet at any one location unless request is affected by engineer in writing.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Install raceway concealed in all areas where required concealment not required in mechanical and electrical rooms, connections to motors, above suspended ceilings, and underfloor spaces.

B. Coordinate installation of conduit in masonry, cabinetry, and building slab work.

C. Underground Raceways: Watertight, including fittings, slope 3 inches per 100 feet downward from building. Install underground marking tape. Bury 6 inches to 8 inches below grade directly above raceway. Seal exterior junction boxes or provide with drainage.

D. Galvanized rigid steel conduit installed in contact with earth shall be wrapped with 2-half laps of 10 mil, all weather, corrosion protection tape. Underground Galvanized rigid steel shall be PVC coated type.

E. Provide GRC (use PGRC) per NEC 344.10 (including 344.10(B)(1))

F. Route all conduits parallel or perpendicular to building lines.

G. Vertical Runs: Straight and plumb.

H. Raceways Running in Groups: Run at same elevation, properly spaced and supported.

I. Install conduit in concrete slab with minimum 2-inch cover. Do not install conduit larger than one inch maximum in concrete slabs unless approved.

J. Do not interfere with placement of concrete re-bar. Place raceway between re-bar layers. Space at least 8-inches on center. Space as far as possible where terminating at same area. Secure raceway, boxes, inserts, etc. by mechanical means prior to pour.

K. Install conduit free with no dents or bruises. Cap ends to prevent entry of foreign materials and moisture.

L. Clean raceway before installation of conductor.

M. Alter conduit routing to avoid obstructions, minimizing crossovers. Avoid use of bends and offsets where possible. Only bend raceway with an approved conduit bending machine or approved hand (hickey) bender.

N. Provide listed expansion complete fittings with grounding jumpers where conduits intersect building expansion joints and for longer runs where conduit expansion may be excessive.

O. Allow minimum of 6 inches clearance at flues, steam pipes, and heat sources.

P. Dissimilar Metals: Avoid contact with pipe or duct runs of other systems.

Q. Lengths and Bends: Maximum number of bends in any run shall be the equivalent of four quarter bends (360 degrees total). Maximum length of any run shall be 300 feet, less 50 feet for each equivalent quarter bend. Provide Junction and pull boxes to meet these limits.

R. Provide entrance seal for all exterior wall, underground, and exterior slab raceway penetrations.

S. Provide conduit seal where shown, detailed, specified (exterior underground junction boxes), and required.

END OF SECTION

SECTION 26 05 33.16

OUTLET, JUNCTION, AND PULLBOXES

PART 1 - GENERAL

1.01 DESCRIPTION OF WORK

A. Provide electrical boxes and fittings as required for a complete, protected, and operable system.

B. Comply with local Codes and NEC as required for Providing electrical boxes and fittings.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI).

1. C73 Series Dimensions of Attachment Plugs and Receptacles

B. National Electrical Manufacturers Association (NEMA)

1. OS 1 Sheet-Steel Outlets Boxes, Device Boxes, Covers, and Box Supports

2. FB 1 Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable

1.03 APPLICABLE REGULATIONS

A. American National Standards Institute (ANSI).

1. C2 National Electrical Safety Code (ANSI/IEEE C2)

B. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

C. Underwriters' Laboratories (UL).

1. UL50 Cabinets and Boxes (ANSI/UL50).

2. UL514 Outlet Boxes and Fittings (ANSI/UL514).

PART 2 - PRODUCTS

2.01 OUTLET BOXES:

A. No smaller than 4-inch, 1-1/2-inches deep box. Provide raised covers where required for surface mounted outlets, plaster rings on flush outlets. Provide tile

rings where flush outlets installed in tile. Concrete type where installed in concrete.

B. Receptacle Outlets and Flush Switch: 4-inch square box, 1-1/2-inches deep, with single or two-gang plaster ring.

C. Match one piece gang boxes to number of devices, install one device per gang. Devices requiring more than one-gang shall be installed in individual boxes matched to device size. Do not exceed 5-gang configuration per row of devices at same location.

D. Provide galvanized steel interior dry location outlet wiring boxes for emt raceway shaped and sized, to conform to each individual location and installation. Provide with factory knockouts in back and sides, and with threaded holes with screws for securing box covers or devices.

E. Provide outlet box accessories as required. Accessories include mounting brackets, wallboard hangers, extension rings, fixture studs, cable clamps and metal straps for supporting outlet boxes. Choice of accessories is Contractor's option.

F. Outlet Box Covers:

1. Flush Mounting: Bevelled, 302 stainless steel, match device installed or full cover where no device installed.

2. Surface Mounting: Bevelled, steel, pressure formed, match device installed or full cover where no device installed.

2.02 WEATHERPROOF / WET LOCATION OUTLET BOXES:

A. Provide corrosion-resistant cast metal weatherproof outlet wiring boxes, shaped and sized, to conform to each individual location and installation. Provide with threaded conduit ends, suitably configured for each application, including face plate gasket and corrosion proof fasteners.

B. Weatherproof boxes shall have smooth sides, gray finish.

C. Boxes used in contact with earth shall be cast iron alloy with gasketed screw cover and water-tight hubs.

D. Weatherproof Plates: Cast metal, gasketed for switches provide spring loaded sealed door(s).

E. Weatherproof Receptacle Outlet Cover: Cast metal, NEMA 3R, In-Use type, with locking tab. Match device configuration. 3 ¼ -inches internal depth. T&B CK series, or approved

2.03 WEATHERPROOF JUNCTION AND PULL BOXES:

A. Provide galvanized sheet steel junction and pull boxes, with screw-on covers; of the type, shape and size, to suit each respective location and installation; with welded seams and equipped with stainless steel nuts, bolts, screws and washers.

2.04 KNOCKOUT CLOSURES:

A. Provide punched-steel knockout closures for steel boxes.

2.05 PULLBOXES

A. Provide sheet metal in interior dry locations for EMT raceway. Provide cast metal in exterior, or damp locations. Type and material shall conform to National Electrical Code, with screw-on cover.

B. Flush Mounted Pullboxes: Provide overlapping covers with flush head screws, finished in light gray enamel.

C. Box volumes shall meet NEC for size and number of entering conduits and cables.

2.06 UNDERGROUND PULLBOXES

A. Underground Pull Boxes: Cast concrete with suitable concrete cover to withhold loads in location installed. Provide heavy-duty traffic cover where installed with vehicle traffic. Cover and box shall not deform and be rated for location installed. Provide drainage and no less than 4 feet compacted gravel below installation. Size, and configuration to match installation. Provide where required, and shown on Drawings.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Match one piece gang boxes to number of devices, install one device per gang. Do not exceed 4-gang configurations per row of devices at same location.

B. Locate outlet boxes flush other than in mechanical rooms, electrical rooms, and above suspended ceilings. Provide insulation behind box to prevent condensation for boxes mounted in exterior walls.

C. Provide insulation behind box for walls with insulation for sound reduction.

D. Coordinate location and mounting heights with built-in units and cabinetry. Outlet mounting height shall be at same level required for equipment served.

E. When mounting receptacle, or voice/voice outlet boxes above bench or counter, mount box to the side (horizontally) for finished receptacle grounding pole at left.

F. Locate pullboxes and junction boxes concealed above suspended ceilings or in electrical rooms, mechanical rooms, or unfinished areas.

G. Support: Provide adequate support of all outlet boxes. Secure boxes independent raceway, by attaching directly to building structure by approved means.

H. Identify each junction and pullbox with system description including branch circuit numbers of enclosed circuits, and voltage.

I. Secure all raceway to entering boxes with approved bushings, and locknuts.

J. Do not mount boxes back-to-back. Boxes on opposite sides of wall shall be separated by at least 3 inches.

K. Maintain sound transmission and fire properties of surface installed. Provide appropriate fire stop and sound stop materials as required to maintain these properties.

L. Provide separate boxes where two voltage systems have equipment at same location. Provide separate boxes for equipment on emergency power system.

END OF SECTION

SECTION 26 05 53

ELECTRICAL IDENTIFICATION

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Permanent Identification of system components.

1.02 APPLICABLE REGULATIONS

A. National Fire Protection Association (NFPA)

1. NFPA 70: National Electrical Code.

1.03 SUBMITTAL

A. Provide sample of each label type.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Phenolic Nameplate:

1. Three -layer, white front and back with black core.
2. Neatly engraved through outer layer to show white characters on black background.
3. Beveled edges, print lettering.
4. Other colors as specified or shown. Use red for fire alarm, or fire sprinkler only.

B. Stenciling and Silk Screening: Printed lettering with enamel or lacquer paints. Legends contrasting with the background on which applied

C. Panelboard Directory Card: Fiberboard typed. Laminate or place in protective cover.

D. Concealed Box Labels: Permanent black ink such as “Sharpie” pen with neat and legible writing. Red permanent ink for fire alarm.

E. Concealed Conductor Labels: Listed white tape wrapped around individual conductor or cable, with permanent black ink with printed lettering

F. Permanent Self-Adhesive Labels: Temperature, moisture and UV resistant. Metallic and reflective background with printed contrasting lettering.

G. Exterior Weather-proof Adhesive Label. Temperature, moisture and UV resistant with printed contrasting lettering.

H. Outlet / Device covers: Clear / transparent tape with matching printed black ink letter / numbering.

I. Power Conductor Labels: Self Extinguishing Tedlar Wrap Around Label Tape. White label with black lettering. B33 type Printer or Approved self-extinguishing label system.

PART 3 - EXECUTION

3.01 EQUIPMENT TO BE IDENTIFIED

A. Motor starters, panels, lighting panels and the disconnecting devices contained therein.

B. Egress/Emergency Lighting

C. Power Conductors (at Panel/Lighting Control Panel/Line Terminal Blocks)

D. Remote Emergency Inverters

E. Disconnects.

F. Control panels, starters, pushbutton stations, pilot lights and other control devices.

G. Transformers.

H. Remote control devices.

I. Conductors at both device and terminal strip terminations for control and instrumentation cables and conductors.

J. Other items as specified, required by NEC, or noted on Drawings

1. Including and not limited to Arc-Flash Hazard (NEC 110.16), Available Fault current (NEC 110.24), protection ratings, etc. Code required markings shall be adhered/affixed/integrated into to equipment from factory with factory labeling/nameplates.

K. Devices in lighting panels and power panels shall be identified on the panelboard directory card.

L. Receptacles and Switches (clear transparent on device cover)

M. Junction Boxes.

N. EV Charging Stations.

3.02 PHENOLIC NAMEPLATES

A. Power panels shall be labeled on the door of the interior with a nameplate. Letters for panels shall be printed and no less than 1/2 inch high.

B. Provide nameplate on switchboard

C. Provide nameplates where specified and as shown.

3.03 POWER CONDUCTOR LABELS

A. Label at emanating location (source)- i.e. home run termination.

B. Applicable to every power source over 70V.

C. Label ~ 1-inch from breaker lug, Relay Lug, Terminal Block. Orient for view toward open readily viewable side.

D. Label each Hot Conductor, Label Neutral Conductor(s) as indicated. Circuit Ground Labeling is not required.

1. Every new conductor home-run at emanating power panel.

    a. Load name shall be used, not circuit number. Refer to description in panel schedule.

    b. Associate/same label Neutral on same circuits

2. Every existing conductor home-run at emanating power panel where re-named if existing.

    a. Load name shall be used, not circuit number. Refer to description in panel schedule.

    b. Existing Muti-Wire Branch Circuits (shared) Neutral- Label as “SHARED” or “MULTI-WIRE” with associated circuits, unique circuit identification required/abbreviations acceptable. If associated circuiting is unknown, label only as “SHARED” or “MULTI-WIRE” without associated circuiting is acceptable.

    c. Associate/same label Neutral on same circuits that are not existing Multi-wire branch circuits.

3. Circuit numbers shall not be indicated, numbers only are not acceptable, load name required for future re-configuration of panel.

3.04 APPLYING IDENTIFICATION

A. Stenciled letters shall be applied by brush or by spraying.

B. Nameplates shall be attached with either adhesive or screws. If adhesive is used, it shall adequately adhere to the surface installed.

3.05 IDENTIFICATION REQUIREMENTS

A. Indicate Voltage for all concealed labels, and for Disconnects, panelboard and switchboard identification.

B. Identification for disconnecting devices contained in panels and motor control centers shall show the equipment name and location by floor, area, and direction to adequately indicate location of load. Do not include Voltage when the Voltage is the same as for the panel or motor control center.

C. Nameplates on disconnect devices located in the area but not part of a panel or motor control center shall have the equipment name, power source identification, and voltage designation. Nameplates for disconnect devices located remotely from the equipment shall also show the equipment location by floor, area, and direction to adequately indicate location of load.

D. All indicators and controls for control panels, starters, and the like shall be labeled, such as (Start, Stop, On, Off, Reset, Fault, etc.).

E. Panelboard directory cards shall list the circuit numbers and show the equipment name and location supplied by the circuits. Equipment locations shall be shown by floor, area and direction, or by room numbers.

F. Device covers to include serving circuit and panelboard name, include at receptacles, switches, outlets, system furniture connections, floor boxes, etc. Apply clear/ translucent adhesive tape with typed black lettering not more than 1/4” in height via use of label maker.

END OF SECTION

SECTION 26 05 60

OVERCURRENT PROTECTIVE DEVICES

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Circuit Breakers.

B. Fuses.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI).

1. C37.16 Preferred Ratings, Related Requirements, and Application Recommendations for Low Voltage Power Circuit Breakers and AC Power Circuit Protectors.

2. C37.17 Trip Devices for AC and General-Purpose DC Low-Voltage Power Circuit Breakers.

3. C37.50 Test Procedure for Low-Voltage AC Power Circuit Breakers Used in Enclosures - Test Procedures.

4. C97.1 Low Voltage Cartridge Fuses 600 Volts or Less.

B. Institute of Electrical and Electronic Engineers, Inc. (IEEE).

1. 20-73 Low Voltage AC Power Circuit Breakers Used in Enclosures: ANSI C37.13.

C. National Electrical Manufacturer's Association (NEMA).

1. FU-1 Low Voltage Cartridge Fuses.

1.03 APPLICABLE REGULATIONS

A. Underwriters' Laboratories (UL).

1. UL 489-72 Molded Case Circuit Breakers and Circuit Breaker Enclosures.

2. UL 198 E Class R Fuses.

3. UL 869 Service Disconnects

4. UL 1066 Standard for Low-Voltage AC and DC Power Circuit Breakers Used in Enclosures

B. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

1.04 QUALITY CONTROL

A. Breakers shall be inherently coordinated with feeder breakers by manufacture.

B. Breakers shall be Eaton to match (E).

PART 2 - PRODUCTS

2.01 CIRCUIT BREAKERS

A. Construction

1. Bolt-on connection to bus.
2. Thermal-magnetic, molded case, with inverse time current overload and instantaneous magnetic tripping.
3. Quick-make, quick-break, with tripped indication clearly shown by breaker handle taking a position between ON and OFF.
4. Multiple phase breakers shall have a common internal trip. Do not use handle ties between single pole breakers.
5. Breaker shall be switch (T) rated.
6. Include locking handle for each breaker provided it does not impede GFI/AFCI or other testing requirements. Utilize for maintenance lock-out tag-out.
7. Where used as service disconnects, breakers shall include locking handle.
8. Fully rated at fault current of panel or switchboard.
9. Inherently coordinated by manufacturer with upstream protection device.

2.02 GFI BRANCH CIRCUIT BREAKERS

A. Meet construction requirements herein.

B. Ground fault protection with integral push-to-test button.

C. Class 1.

D. Adjustable setting pickup from 0.03 to 30 Amps.

E. Adjustable time delay from instantaneous to 2.0 seconds.

2.03 TESTING

A. By Manufacturer at factory. Timed thermal trip test and timed magnetic trip test.

B. As required by local authority.

C. Submit test results to Engineer upon request.

2.04 FUSES

A. Feeder, Branch Circuit and Service Entrance Fuses: 600 amperes and below, UL Class J or RK1 current limiting type, 600 volt 200,000 ampere interrupting capacity.

B. Motor and Inductive Circuit Fuses: UL class RK5 time delay current limiting type, 600 volt, 200,000 ampere interrupting capacity.

C. Control Circuit Fuses: UL Class J or R current, limiting type, 600V.

D. Manufacturer: Bussmann, or approved

PART 3 - EXECUTION

3.01 CIRCUIT BREAKER INSTALLATION

A. Label each breaker located in switchboard or separate enclosure to indicate load served.

B. Adjust settings on breakers to operate properly under actual field conditions and to provide selective system coordination.

C. Post breaker settings in electrical rooms, alongside one-line.

D. Torque breakers to bus per manufacturer's requirements and installation procedures.

3.02 FUSE INSTALLATION

A. Label each switch to indicate type and rating of fuse installed.

B. All fuses shall be selected to provide selective system coordination.

C. Provide the greater of 10%, or not less than 3 spare fuses of each size, and rating used.

END OF SECTION

SECTION 26 05 83

WIRE CONNECTIONS

GENERAL

1.01 WORK INCLUDED

A. Wires Connectors

1.02 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

B. Underwriters' Laboratories, Inc (UL

1. UL 486A through UL 486E

C. American National Standards Institute (ANSI)

1. ANSI/UL 467

PRODUCTS

1.03 TWIST-ON CONNECTOR

A. Pressure-type wound spring twist on connector.

B. Solderless pressure connectors.

C. Shell rating of 105 degrees C.

D. "Push-On" or "punch" type connectors not permitted.

1.04 COMPRESSION ADAPTER

A. Dual rated for use with both aluminum and copper cable conductors.

B. Diameter and ampacity as current carrying equivalent copper wire.

C. Pre-filled with approved joint compound

D. Connectors shall be clearly marked with Catalog Number, wire size and color-coded die index number.

E. Burndy "Hyplug" type AYP or equal by T&B, or approved.

1.05 TERMINAL, CRIMP TYPE

A. Flat; fork tongue, or flat circular matched to terminal size.

B. Color coded to wire size.

C. T&B “Sta-Kon”, or approved.

1.06 WP COATING

A. Liquid

B. For use as an outer seal on vinyl tape splice, fast- drying, suitable for use for direct burial and moisture protection.

C. 3M Scotchkote Electrical Coating FD, or approved

1.07 AS SHOWN ON DRAWING.

EXECUTION

1.08 INSTALLATION

A. Provide Twist-On Connectors at taps and splices for conductors no larger than #10 AWG. Provide only in approved junction and outlet boxes.

B. Provide Compression Adapters for terminating a single conductor into mechanical connectors such as a circuit breaker or set screw lugs. Provide only where required for AL/CU transitions or where lugs require adapters.

C. Provide Crimp terminal at all Control voltage terminal blocks, unless otherwise recommended by manufacturer.

D. Do not nick conductor when stripping insulation. No “ringing”

E. Conductor and cable shall not be reduced at the terminal for connections.

F. Connectors shall be approved and listed for the purpose used.

G. Wrap all twist-on connectors with listed tape to maintain equivalent insulation of wire.

1. Exterior, Underground, and WP connections shall include 2-coats of WP Coating.

H. Remove any obstructions on connection to maintain continuity prior to installation of connectors, such as paint, dirt, and construction materials.

I. Copper conductors can be terminated in approved compression or mechanical connector, including set screws.

J. Provide slack at equipment to allow for a neat termination, access to conductors, and ability to repair or replace equipment.

END OF SECTION

SECTION 26 27 26

WIRING DEVICES

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Wall Switches.

B. Receptacles.

C. Ground Fault Receptacles.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI).

1. C73 Series Dimensions of Attachment Plugs and Receptacles.

B. National Electrical Manufacturer's Association (NEMA).

1. WD 6 Wiring Devices- Dimensional Requirements
2. WD 1 General Color Requirements for Wiring Devices.

C. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

D. Underwriters' Laboratory (UL).

1. UL-20 Standard for Snap Switches.
2. UL 498 Attachment Plugs and Receptacles
3. UL 467 Grounding and Bonding Equipment
4. UL 514D Cover Plates for Flush-Mounted Wiring Devices
5. 2006 UL 943 Safety for Ground-Fault Circuit-Interrupters

1.03 QUALITY ASSURANCE

A. Provide type 5362 receptacles.

B. Acceptable Manufacturers: Hubbell, Pass and Seymour, Arrow-Hart, Leviton, or approved

PART 2 - PRODUCTS

2.01 MATERIALS

A. Switches: 120/277 Volt. AC Quiet, slow make, slow break design, toggle style handle, with totally enclosed case, 20 Ampere, specification grade. Provide matching two-pole, three-way and four-way switches.

B. Receptacles: 20 Ampere (unless otherwise indicated), 125 Volts (unless otherwise shown), duplex, polarized, full gang size, specification grade, separate ground terminal, 20 Amp. break-off tab for split circuit wiring.

C. Ground Fault (GFI) Receptacles: 20 Ampere, specification grade duplex receptacle with integral ground fault circuit interrupter. LED operation indicator. Test and reset buttons. End of life protection- GFI component failure results in no power delivered to equipment (2006 UL 943).

D. Wall Plates: 302 Stainless Steel, Match device configuration.

E. Colors:

1. Gray Receptacles, Gray Switches: In finished areas that are not dark brown in color.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Do not use back wiring wells, terminate conductors on mechanical screw terminals.

B. Do not use GFI-Feed through function, GFI receptacles with test, reset, etc. required at each location where GFI is required.

C. Provide wiring devices as shown.

D. Install devices plumb and consistent with building lines. Wall Plates shall make contact on four corners and shall fit flush with device.

E. Devices to include same configuration outlet box, cover, wall plate and other necessary installation materials for a complete operating circuit.

F. Mount switches 42 inches (to center line of faceplate) above floor except as otherwise noted on the Drawings.

G. Coordinate mounting locations with details.

H. Mount receptacles vertically at 15 inches (to bottom of faceplate) above finished floor, with grounding pole at bottom.

I. Coordinate receptacle height with benches and counters.

J. When mounting receptacle above bench or counter, mount receptacle to the side with grounding pole at left.

K. Grounding: Install a separate bare conductor between the receptacle strap grounding (green) screw and a screw into the outlet box. Self-grounding strap not approved as grounding means.

END OF SECTION

SECTION 26 27 30

HEATING EQUIPMENT

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Provide electric space heating equipment and control as shown on drawings and required herein.

B. Verify all heater size, type, location, control, and control location with Div. 15 and Mechanical

1.02 REQUIREMENTS

A. UL listed equipment

B. NEC compliant installation

C. Thermal safety shutdown

PART 2 - PRODUCTS

2.01 WALL HEATERS

A. Wall heaters shall be the size and voltage as shown with integral thermostat, integral timer recessed type. White in color coordinated with civil.

B. Additional requirements as shown on Drawings.

C. Acceptable manufacturer: Finished Area type: Q-Mark or as shown on Drawings.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Install in accordance to code and manufacturer's requirements and instructions.

3.02 DISCONNECTS

A. Provide disconnects as required for each heater. Lockable breakers at panel may be used if approved. Integral locking tab by panelboard manufacturer required.

3.03 WARRANTY

A. Provide 2 year warranty on heaters.

END OF SECTION

SECTION 26 29 12

DISCONNECTS & MANUAL STARTERS

PART 1 - GENERAL

1.01 WORK INCLUDED

A. Provide motor disconnects as shown, and as required by Codes.

B. Provide circuit disconnects as shown, and as required by Codes.

C. Provide manual motor starters for single phase motors below one horsepower where disconnect is shown as required by Codes.

D. Disconnects to include adequate support, required hardware, and accessories for complete functional installation.

1.02 APPLICABLE REGULATION

A. Conform to National Electrical Code and inspection authority.

B. Provide disconnects rated for the location installed, as required by National Electrical Code, as shown, and as indicated herein.

1.03 REFERENCE STANDARDS

A. Underwriters' Laboratory (UL).

1. UL-98 Enclosed Switches.

B. National Electrical Manufacturer's Association (NEMA).

1. NEMA KS-1 Enclosed Switches.

PART 2 - PRODUCTS

2.01 MANUAL MOTOR STARTER

A. Toggle horsepower rated, switch with thermal overload and pilot light.

B. Switch tab for locking switch in “OFF” Position.

2.02 DISCONNECT

A. Motor and circuit disconnects shall have a UL label.

B. Construction: Dry, Indoor Locations shall be not less than NEMA 1. Enclosures for outdoor, or wet locations shall be not less than NEMA 3R. Rated at 600 Volt.

Heavy duty, quick make, quick break. Number of poles and ampacity as noted or required by Code. Fusible with dual element fuses where shown. Short circuit rating sufficient to withstand the available fault current. Solid ground, solid neutral.

C. Compression lugs or set-screw lugs approved for use with copper wire.

D. ON/OFF Positions clearly marked.

E. Lockable in "OFF" position.

F. Interlock:

1. Prevents switch from being opened when "ON."
2. Prevents closing switch when cover is open.
3. Provide defeater to permit authorized personnel to open door and inspect switch when "ON," or operate with cover open.

PART 3 - EXECUTION

3.01 INSTALLATION

1. Install motor and circuit disconnects as recommended by manufacturer, required by Code, required by UL, and where shown.
2. Maintain Code clearances and access.
3. Provide Manual Motor Starter where shown for single phase motors rated less than one horsepower. Manual Motor Starter is not required for motors with integral thermal overload protection. Provide Switch with locking tab for motors with integral thermal overload protection. Provide Manual Motor Starters as required for single phase motors without integral or other protection, otherwise provide Disconnect.
4. Provide a phenolic nameplate on each disconnect identifying the equipment item served.
5. Independently support disconnects. Do not mount disconnects on vibrating, HVAC, or Plumbing equipment.
6. Install heaters correlated with full load current of motors provided.
7. Set overload devices according to measured current of motors provided.

END OF SECTION

SECTION 26 05 33

VARIABLE FREQUENCY DRIVE

PART 1 - GENERAL

1.01 SUMMARY

A. The Variable Frequency Drive (VFD) system shall contain all components required to meet the performance, protection, safety and certification criteria of this specification.

1.02 REFERENCES

A. National Fire Protection Association - NFPA 70 - US National Electrical Code.

B. National Electrical Manufacturers Association - NEMA 250 - Enclosures for Electrical Equipment.

C. Underwriters Laboratory Inc. – UL 508.

D. International Electrical Code - IEC 146.

E. Institute of Electrical and Electronics Engineers, Inc. - IEEE 519 - IEEE Standard Practices and Requirements for Harmonic Control in Electrical Power Systems.

F. NEMA ICS 3.1 - Safety standards for Construction and Guide for Selection, Installation and Operation of Adjustable Speed Drive Systems.

1.03 SUBMITTALS

A. Submit under provisions of Division 01, and Section 26 00 26

B. Shop Drawings – Acceptance

1. Elevation Drawings: Include dimensional information and conduit routing locations.

2. Unit Descriptions: Include amperage ratings, enclosure ratings, fault ratings, nameplate information, etc. as required for approval.

3. Wiring Diagrams:

a. Power Diagram: Include amperage ratings, circuit breaker frame sizes, circuit breaker continuous amp ratings, etc. as required for approval.

b. Control Diagram: Include disconnect devices, pilot devices, etc.

4. Major components list.

C. Product Data Sheets

1. VFD and Operator Interface publications.

2. Data sheets and publications on all major components including but not limited to the following:

a. Contactors

b. Circuit breaker and fuse (power and control)

c. Control power transformers

d. Pilot devices

e. Relays/Timers

D. Test procedures shall be per the manufacturer's standards.

1.04 QUALITY ASSURANCE

A. Qualifications:

1. Manufacturers:

a. The VFD and all associated optional equipment shall be UL listed or recognized.

b. The VFD shall contain a UL label attached on the inside of the enclosure cabinet.

2. Suppliers:

a. All inspection and testing procedures shall be developed and controlled under the guidelines of the Supplier's quality system and must be registered to ISO 9001 and regularly reviewed and audited by a third party registrar.

b. The VFD shall be factory pre-wired, assembled and tested as a complete package.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Contractor shall coordinate the shipping of equipment with the manufacturer.

B. Contractor shall store the equipment in a clean and dry space at an ambient temperature range of -25 °C to 55 °C (-13 °F to 130 °F).

C. The contractor shall protect the units from dirt, water, construction debris and traffic.

1.06 WARRANTY

A. The manufacturer shall provide their standard parts warranty for eighteen (18) months from the date of shipment or twelve (12) months from the date of being energized, whichever occurs first.

B. This warranty applies to variable frequency drive systems.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Allen-Bradley – PowerFlex 70 Services VFD, or Allen Bradley controls compatible approved.

2.02 VARIABLE FREQUENCY DRIVE UNIT

A. Features

1. Certifications

a. Listed to UL508C

b. In conformity with EMC Directive, Low Voltage Directive, and applicable IEC and EN standards. Drive shall have an internal EMC filter to meet second environment EMC requirements without additional components where applicable.

c. Designed to meet NEMA ICS 3.1, NEMA 250, CAN/CSA C22 No.14-M91, and IEC 146.

d. Compliance with NFPA 70, IEEE 519 recommendations for harmonic control (with appropriate filters if required), and SEMI F47 certification

2. Hardware

a. Utilize diode or full bridge rectifier on the input.

b. DC bus inductor on ratings 7.5HP (5.5kW) or greater.

c. Switching logic power supply operating from the DC bus..

d. Incorporate Metal Oxide Varistor (MOV) phase-to-phase and phase-to-ground surge protection where applicable.

e. Common mode capacitors on units above 2 HP, 480V with optional jumpers for ungrounded systems.

f. Gold-plated plug-in connections on printed circuit boards.

g. Microprocessor-based inverter logic, isolated from power circuits.

h. Inverter section shall use latest generation IGBT devices with nominal rise times of 200ns or longer.

i. No commutation capacitors required.

j. Common control connection for all ratings.

k. Device Peripheral Interface (DPI) for connection to common options.

l. Status LEDs for drive condition and communications visible through the cover.

m. Enclosures available as IP20/Type 1, IP66/NEMA 4X/12 (indoor), and flange mount options.

3. Control Logic

a. Ability to operate with the motor disconnected.

b. Provide a controlled shut down, when properly fused, without component failure in the event of output phase-to-phase or phase-to-ground short circuit. Provide annunciation of the fault condition.

c. Provide multiple programmable stop modes including Ramp, Coast, DC-Brake, and Ramp-to-Hold.

d. Provide multiple acceleration and deceleration rates adjustable from 0 to 3600 seconds.

e. Adjustable output frequency up to 500Hz.

f. Selectable V/Hz, Sensorless Vector, and full Vector Control with optional encoder feedback.

g. Provide the ability to function as stand-alone or integrate into supervisory control.

h. Programmable digital inputs/outputs and analog inputs/outputs.

i. Full text multi-line LCD display (HIM) supporting multiple languages.

j. Status monitoring includes fault, alarm, direction, speed/frequency, and run conditions.

k. Self-tuning startup routines (S.M.A.R.T. Start and Assisted Start Up).

l. Auto Restart feature with up to nine automatic fault resets.

m. Internal Process PID regulator with integral, proportional, and derivative control.

n. Multiple frequency skip bands to avoid mechanical resonance.

o. Integrated timer functions, scaling, and signal conditioning for I/O.

4. Current Limit

a. Programmable current limit from approximately 110% to 150% of rated current for normal or heavy duty ratings.

b. Current limit active in all drive states: accelerating, constant speed, and decelerating.

c. PI regulation ensures smooth transitions in and out of current limit.

5. Acceleration / Deceleration

a. Independent Accel/Decel settings adjustable from 0.0 to 3600.0 seconds.

b. Remotely selectable accel/decel settings through digital inputs.

c. S-Curve acceleration/deceleration profiles are adjustable.

6. Speed Profiles

a. Capable of producing linear, fan/pump, or custom V/Hz patterns.

b. Vector Control mode provides a wide speed range. Sensorless Vector: 120:1 speed range; Closed Loop Vector: 1000:1 speed range.

7. Adjustments

a. A digital interface (HIM) for all setup, operation, and adjustment.

b. All adjustments stored in nonvolatile memory.

c. No potentiometer adjustments required.

d. Nonvolatile memory for factory default values.

e. Software tools available for trending, diagnostics, and offline programming.

8. Process PID Control

a. Internal PID regulator with proportional, integral, and derivative adjustments as well as error inversion and anti-windup protection.

b. Loss of feedback protection and signal scaling included.

c. Capable of “ride-through” power dips, maintaining control during short outages.

9. Skip Frequencies

a. Up to three adjustable skip frequency bands to avoid resonance.

10. Fault Reset / Run

a. Up to nine automatic fault resets before requiring manual restart.

b. Automatic mode not applicable to ground faults or other severe internal faults.

c. Time between restarts adjustable from 0.5 to 30 seconds.

11. Run on Power Up

a. User-programmable restart after restoration of power following outages.

12. Fault Memory

a. Last four faults stored with time stamps.

b. Additional status conditions, voltage, current, and operational data stored at time of fault.

c. Power-up markers recorded for analyzing startup conditions.

d. Alarm codes stored and time-stamped for troubleshooting.

13. Overload Protection

a. UL listed Class 10 motor overload protection compliant with NEC Article 430.

b. Overload protection adjustable and speed sensitive.

14. Flying Start

a. Capable of determining the motor's speed and direction if spinning and "catching" the motor at rotating speed.

15. Inputs and Outputs

a. Configurable digital and analog I/O, isolated from drive logic.

b. No jumpers or switches required for basic digital input configuration.

c. Analog inputs (0-10 V / 4-20 mA) and analog output (0-10 V or 4-20 mA) fully programmable.

d. Multiple reference signals: Analog inputs, Preset Speeds, Digital MOP, HIM, Encoder feedback, Network references.

16. Reference Signals

a. Capable of using digital MOP, jog, HIM, analog input signals, preset speeds, network references, and encoder input.

b. Capable of sensing loss of reference and programmable responses (fault, alarm, maintain last speed, etc.).

17. Loss of Reference

a. Programmable action on reference loss: Fault, Alarm, Maintain Speed, or Run at Preset Speed.

18. Metering

a. Human Interface Module provides output current, voltage, power, DC bus voltage, frequency, heatsink temperature, elapsed run time, and fault history.

19. Faults

a. The drive provides annunciation of faults including power loss, undervoltage, overvoltage, motor overload, heat sink over-temperature, and phase-to-phase/phase-to-ground faults.

20. Predictive Diagnostics

a. Parameters for relay output life, fan life, and bearing life based on operating hours.

2.03 VFD PACKAGED SYSTEM

A. Features

1. Ratings

   a. Input Voltages

      1) 200-240 VAC ±10%

      2) 380-480 VAC ±10%

      3) 500 VAC –13.6% to 600 VAC +10%, three-phase, 50/60 Hz

   b. Power Factor: ≥0.95 displacement at rated load/speed.

   c. Efficiency: ~97% at full load and nominal line voltage.

   d. Operating Temp

      1) IP20/Type 1: 0°C to 50°C (32°F to 122°F) without derating.

      2) IP66/Type 4X/12: 0°C to 40°C (32°F to 104°F) without derating

   e. Humidity: 5% to 95% non-condensing

   f. Altitude: Up to 1000 m (3,300 ft) without derating.

   g. The VFD shall be rated for not less than 50 HP at 480 VAC, three-phase.

2. Overload Capability

   a. Normal Duty: 110% for 1 min, 150% for 3 sec.

   b. Heavy Duty: 150% for 1 min, 200% for 3 seconds

3. Input Line Reactor

   a. Provide an appropriately sized input line reactor (e.g., 3% or 5% impedance) as recommended by the manufacturer to reduce line harmonics and protect the drive's input section

   b. The reactor shall be factory-installed or provided as a separate component per manufacturer's instructions

   c. Show version/data with Submittals.

4. Auto Reset/Run

a. Automatic reset and restart for certain recoverable faults.

5. Ride-Through

a. Adjusts output during short power dips based on inertia and settings.

6. Run on Power Up

a. Programmable restart after power restoration.

7. Communications

a. DPI port supports up to four peripherals.

b. Optional communication interfaces: DeviceNet, EtherNet/IP, ControlNet, Remote I/O, etc.

c. The VFD shall integrate seamlessly with the existing Allen-Bradley control system, including Allen-Bradley PanelView Plus 7 operator interfaces, via EtherNet/IP or other specified Allen-Bradley compatible network protocols

d. Interface with (E) Human Interface Module (HIM).

e. See Drawing for additional requirements

8. Enclosure

a. NEMA/UL IP66/Type 4X/12

b. Painted per manufacturer standard.

c. Provide cable entry/exit.

d. Nameplate indicating SCCR and other ratings.

9. Drive Enclosure Input Disconnect (If required)

a. Main disconnect device

b. Externally operable, lockable handle

10. Auxiliary Relays

a. Two Form C (N.O./N.C.) output relays, rated per PF70 specs (250VAC, etc.).

11. Local Control Interface

a. Terminals rated for the required control voltage

b. Provide terminals for start, stop, enable, external fault, speed select.

12. Output Filtering

a. Drive includes a software algorithm to limit reflected wave and hardware to limit peak motor voltages.

b. See Drawings, VFD Cable Required.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify that location is ready to receive equipment.

B. Verify that the building environment can be maintained within the service conditions required by the manufacturer of the VFD.

3.02 INSTALLATION

A. Installation shall be in compliance with all manufacturer requirements, instructions and drawings.

3.03 START-UP SERVICE

A. At a minimum, the start-up service shall include:

1. Perform pre-Power Check

2. Megger Motor Resistances: Phase-to-Phase and Phase-to-Ground

3. Verify system grounding per manufacturer's specifications

4. Verify power and signal grounds

5. Check connections

6. Check environment

B. Drive Power-up and Commissioning:

1. Measure Incoming Power Phase-to-Phase and Phase-to-Ground

2. Measure DC Bus Voltage

3. Measure AC Current Unloaded and Loaded

4. Measure Output Voltage Phase-to-Phase and Phase-to-Ground

5. Verify input reference signal

C. Measurements shall be recorded.

D. Drive shall be tuned for system operation.

E. Drive parameter listing shall be submitted.

3.04 PROGRAMMING

A. See Drawings for additional requirements

END OF SECTION

SECTION 27 05 01

CONTROLS CABLING

PART 1 - GENERAL

WORK INCLUDED

A. Provide Control cabling as required for complete, and operable systems. Control conductor cabling shall be per system manufacturer, and adhere to requirements herein. Additional requirements may be required by manufacturer, or by other sections/drawings. Herein includes base requirements. Follow installation, operation, and maintenance requirements

B. In general, the Project work includes installation of new controls cable, terminations, face plates, outlets, distribution frame hardware, cable terminations, labeling, grounding and bonding, firestopping, and conformance testing of the installation in addition to other requirements specified herein.

1.02 SUBMITTALS

A. See Execution Section herein.

1.03 APPLICABLE STANDARDS

A. Work shall be provided per Code. Provide per manufacturers requirements and recommendations. Materials shall be UL approved or listed by accredited and approved listing agency. Equipment shall be labeled where a UL standard and/or test exists.

B. The latest edition of referenced standards from the latest available draft in the case of proposed standards; shall be utilized.

1. Underwriters Laboratories (UL)

    a. UL 444 Communications Cables

    b. UL 1863 Standard for Communications-Circuit Accessories

    c. UL 2269 Optical Fiber/Communications/Signaling/Coaxial Cable Outlet Boxes

2. Underwriters' Laboratories of Canada

    a. ULC-S139 Fire Test for Evaluation of Integrity of Electrical Cables

3. American National Standards Institute (ANSI), Telecommunications Industry Association (TIA), Electronic Industries Alliance (EIA)

a. ANSI/TIA/EIA-492 AAA Multi Mode Optical Fiber

b. ANSI/TIA/EIA-568-A Commercial Building Wiring Standard

c. ANSI/TIA/EIA-568-B Commercial Building Telecommunications Cabling Standard

d. ANSI/TIA/EIA-854 A Full Duplex Ethernet Specification for 1000Mbis/s (1000BASE-TX) Operation Over Category 6 Balanced Twisted-Pair Cabling.

e. ANSI/TIA/EIA 607 Commercial Building Grounding and Bonding Requirements for Telecommunications.

4. International Standards Organization (ISO), International Electrotechnical Commission (IEC)

   a. ISO/IEC 11801 Generic Cabling for Customer Premises

   b. ISO/IEC 60603-7 RJ45 Connector (8-Pin)

   c. ISO/IEC 60793-2 Optical Fibers

5. Electrical Code

6. National Electrical Manufacturers Association (NEMA)

   a. NEMA WC-63 Performance Standards for Twisted Pair Premise Voice and Data Communications Cable

   b. NEMA WC- 66 Performance Standard for Category 6 and Category 7 100 Ohm Shielded and Unshielded Twisted Pair Cables

7. ANSI, Insulated Cable Engineers Association (ICEA)

   a. ANSI/ICEA S-90-661 Category 3, 5, & 5e Individually Unshielded Twisted Pair Indoor Cable for Use In General Purpose and LAN Communication Wiring Systems.

C. Applicable Testing Standards: Testing of individual components and channel shall be conducted in accordance with the following standards:

1. ASTM D 4566-94, Standard Test Methods for Electrical Performance Properties of Insulation and Jackets for Telecommunications Wire and Cable

2. ANSI/TIA/EIA-492 AAA, Multi Mode Optical Fiber

3. ANSI/TIA/EIA-568-A, Commercial Building Telecommunications Standard
4. ANSI/TIA/EIA-568-A-1, Propagation Delay and Delay Skew Specifications for 100 W 4-pair cable
5. ANSI/TIA/EIA-568-A-2, Corrections and Additions to TIA/EIA-568-A
6. ANSI/TIA/EIA-568-A-4, Production Modular NEXT Loss Test Method and Requirements for Unshielded Twisted Pair Cabling
7. ANSI/TIA/EIA-568-A-5, Transmission Performance Specifications for 4-pair 100 W Category 5e Cabling
8. ANSI/TIA/EIA-TSB 67, Transmission Performance Specifications for Field Testing of Unshielded Twisted Pair Cabling Systems
9. ISO/IEC 11801

PART 2 - PRODUCTS

2.01 TERMINATION HARDWARE AND ACCESSORIES

A. See Wire Connections for #14 AWG, additional requirements.

1. Screw Terminals- Compression adapter Sta-Kon, or approved system
2. Other types as compatible with Device/Controller connection point (i.e. compression tabs, plugs, etc.
3. Full insulation such that at no point should conductor be exposed when properly seated in connection point. Insulation stripping not to exceed connection system beyond terminal block/connection mechanism.

B. Shielded cabling shall be connected to communications ground bus system

C. As Required by Control System manufacturer

D. Raceway required throughout.

2.02 COPPER DATA/COMMUNICATIONS CABLE

A. Category 6

B. Meets ULC S139

C. Copper twisted 4-pair, unshielded

D. Plenum rated where routed through plenums

E. Impregnated blue, or per (E), facility standard.

F. Acceptable cable manufacturers: CommScope, LANmark, Beldon, Mohawk, Corning, or approved.

PART 3 - EXECUTION

3.01 GENERAL

A. Provide conductors as required Allen Bradly compatible.

B. It is the intent of the Drawings and Specifications that systems be complete, fully tested and ready for operation.

C. Cables systems are shown and presented as a performance specification. Submit Shop Drawings, including connection points, terminals, risers, and the like for review. Contractor shall provide detailed drawings showing cabling and riser information associated with specific control system.

D. Contractor shall provide cable support hardware in every location.

E. Bundle cable groups utilizing equipment specified. Do not staple cables in any location.

F. Provide raceway throughout.

G. The CAT 6 cable system shall follow ANSI/TIA/EIA and BICSI guidelines and shall consist of cable and connecting hardware manufactured by Ortronics or approved. For direct assistance in interpreting telecommunications standards, utilize the services of a Registered Communications Distribution Designer (RCDD) certified by the Building Industry Consulting Service International (BICSI).

H. Submit labeling information. Re-apply labeling that has not been coordinated with Controls, Controls Manufacturer, and the Govt., or that is not permanent.

I. Coordinate all cabinetry locations, and cable routes.

J. Route all cables in a neat a professional manor, concealed in raceways. Raceways/Cabling shall not block or impede access, spaces, walkways, or other areas indented for a different function. Route parallel to building lines, and group in same areas.

K. Secure all cable, loose or “sagging” cable across spans is not acceptable.

L. Route cable between origination and termination points between the shortest distance feasible through the raceway systems. Observe particular attention to connections further distances from the controllers.

M. Do not pinch or staple cables in any locations. Pinched or stapled cables are not acceptable.

N. Route cables and maintain bending radius of cables, and raceway as required by cable manufacturer and applicable standards.

O. Do not splice cables.

3.02 GROUNDING

A. Ground equipment racks/enclosures/cable shields utilizing ground bus bars and grounding conductor not less than as required by Code and standards.

B. Utilize compression lugs at ground bus locations.

C. Bond to the following:

1. As shown on Drawings
2. Controller Enclosures
3. Telecommunications Panelboards/Equipment
4. Telecommunications Racks/Mounting Hardware
5. Distribution grounding system

3.03 TESTING

A. CAT 6 Cables

1. Provide testing for performance with IEEE Category 6 requirements, and ANSI/TIA/EIA performance requirements.
2. Visually inspect cables and packaging to detect cable damage incurred during shipping and transport. Return visibly damaged items to the manufacturer.
3. Conduct cable testing as described below upon completion of installation. Test fully completed systems only. Sample or Piecemeal testing is not acceptable, except by prior written approval from the Architect and Owner.
4. After terminating both ends, but before any cross connects are installed, test all voice and data stations cables for attenuation and for near-end cross talk (NEXT) to CAT 6. Test all UTP backbone, distribution and patch cable for cable pair/conductor continuity, ground fault, proper cross-connect, shorts, loose connectors, and crossed pairs.

5. Remove all defective cables from pathways system. Do not abandon cables in place.

6. Maintain and complete a comprehensive typed testing log showing testing results of all cables. Log shall include cable identification as required herein and shown. Indicate cables that were replaced as a result of testing. Submit Log for Owner with O&M manuals. Execution of testing log does not alive Contractor from providing functional equipment as required herein regardless of testing results.

B. Control Cables (#14 & shielded)

1. Continuity Test (full length, test installed cable)

   a. Connection present per standard Klien, Fluke, or approved Digital Multi-Meter noise sound emitted threshold Ohm Setting of sound made when connection present.

2. Resistance Test (full length of installed cable, bench test, uncoiled conductor)

   a. Not greater than 5% Voltage Drop

   b. Increase cable gauge to maintain these limits

3. Shield Continuity Test (full length of installed cable, bench test, uncoiled conductor)

   a. Resistance meets manufacturer requirements for length required (integrity maintained)

4. Additional as specified by Controls, and/or Orenco

C. See 26 00 10 for additional requirements including additional procedures to above.

3.04 ACCEPTANCE

A. Upon receipt of the Contractor's log of cable testing, the COR will review the installation and may request a retest using Contractor equipment and labor, of up to 10% of the cable/wires installed. Inconsistent results are not acceptable.

END OF SECTION